

Recurrent Spiking Predictive Coding for Sparse Representations

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Introduction

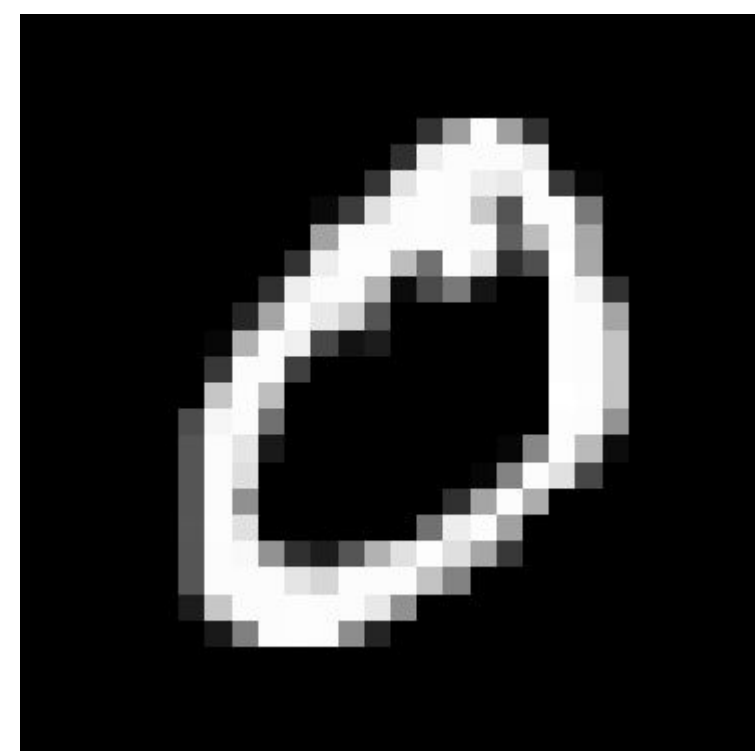
Energy usage is a major issue in AI applications

- Biological systems like the brain are highly energy efficient
- Predictive coding is a biologically inspired alternative to feedforward deep learning via backpropagation

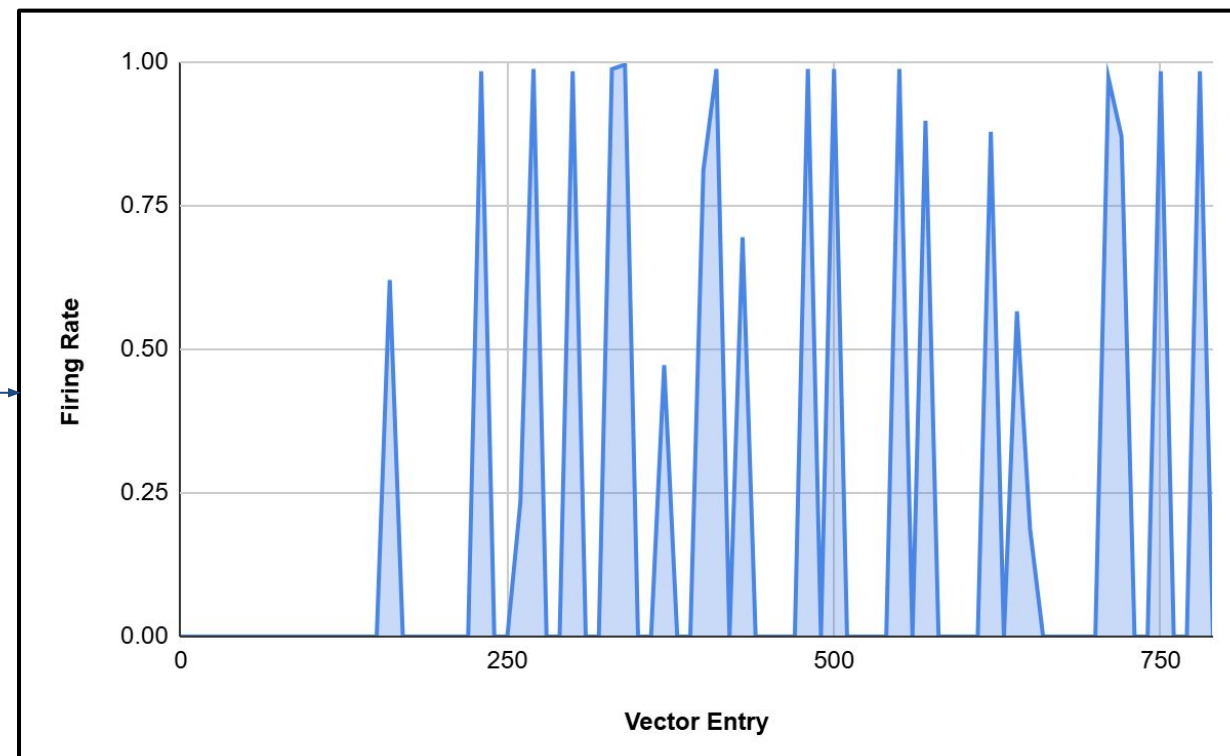
Spiking Dynamics

- Firing rates approximate input vector values
- Input current integrated over a discrete time window

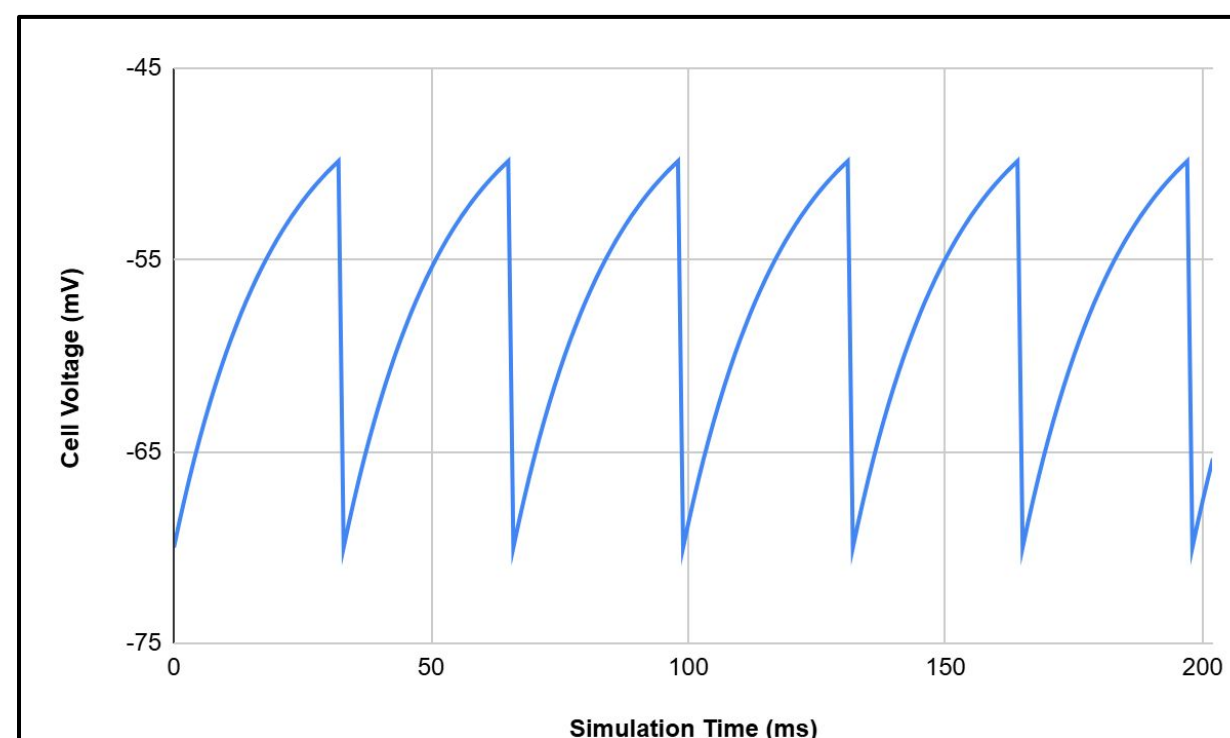
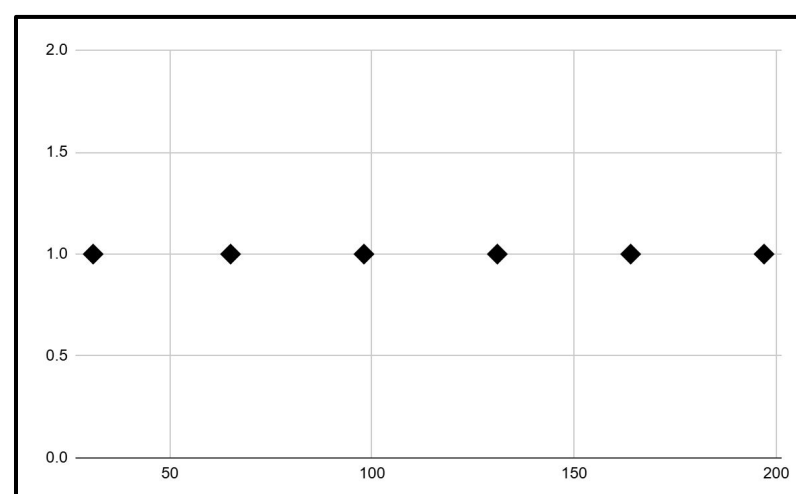
Input



Spike Frequency

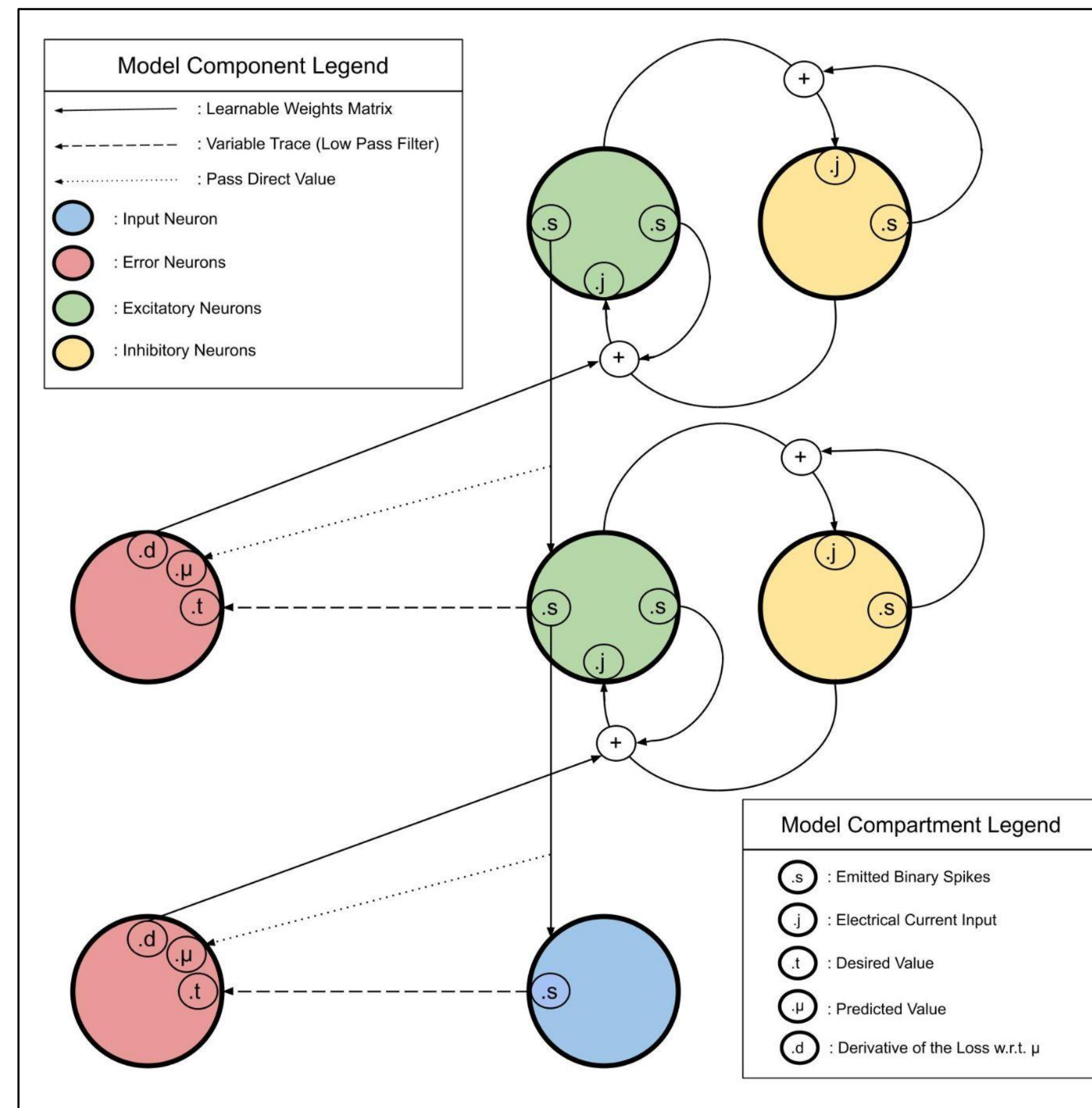


- Governed by Ordinary Differential Equations
- Neurons can be tuned to alter their spiking dynamics with adaptive threshold and refractory periods

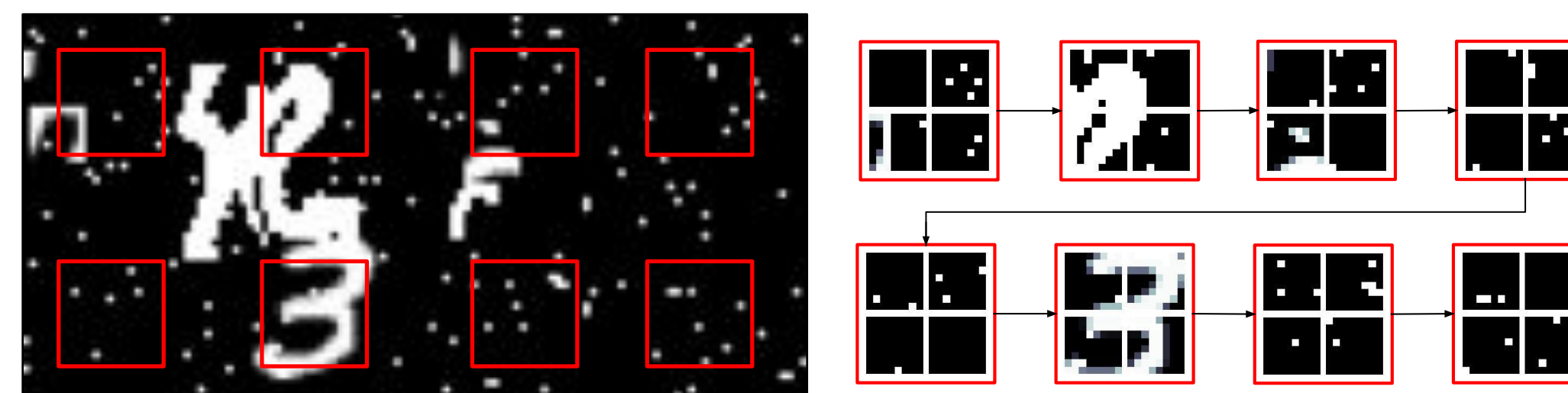


$$\tau_m \frac{dv}{dt} = (v_{\text{rest}} - v) + jR$$

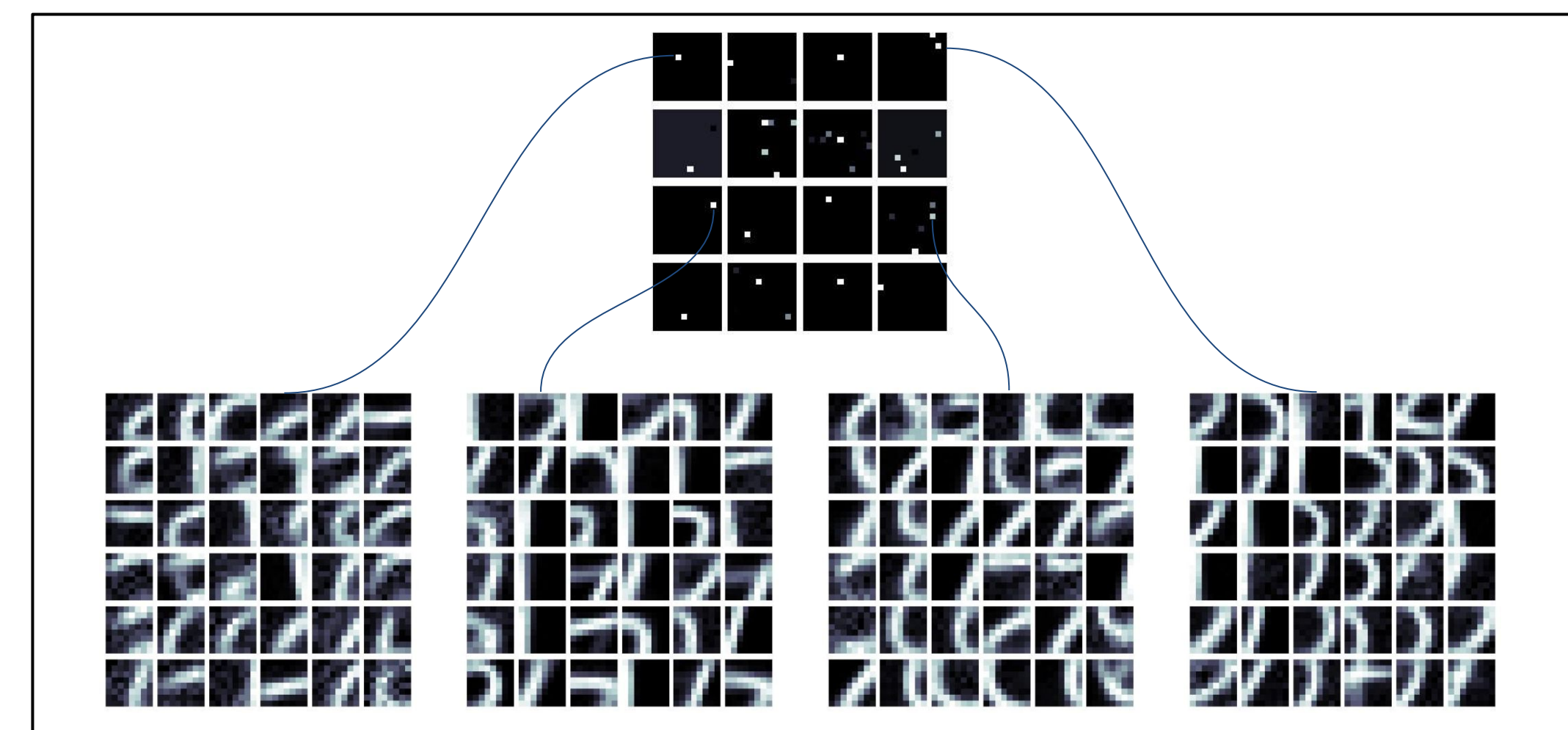
Architecture



- Our network utilizes recurrence and populations of inhibitory and excitatory neurons in an 80:20 ratio
- Bottom up, top down, and recurrent signals facilitate learning via mismatches between predictions computed by error cells
- We study glimpses over a trajectory to gather network statistics



Results



- Sparse receptive fields
- Emerging control structures in the second layer

Conclusions

The emergence of receptive fields and sparse control structures in the upper layer demonstrates our network's ability to learn sparse, meaningful representations. These results support biologically inspired machine learning methods as a promising direction for future research, especially in the area of sparse representations.

Future work

- Classification to demonstrate interpretability
- Comparisons to neuronal data and scene saliency

Acknowledgements

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